PHYSICAL CHEMISTRY CUMULATIVE EXAM

April 5, 1997

An interesting radioactive decay is electron capture. In this reaction, a proton in the atomic nucleus reacts with an electron of the atom, giving a neutron. In the process, the atomic number decreases by one while the atomic mass is approximately unchanged. An important example is the decay of cobalt-57 to iron-57 and γ radiation. There have been proposals to use this strong γ radiation as a nuclear weapon (cobalt bomb). The highly monochromatic γ radiation is also used in Mössbauer spectroscopy.

The reaction takes place only if an electron is within the nucleus, as the nuclear force responsible for the reaction has an extremely short range. This mean that practically only the 1s electrons react.

(a) Estimate the probability that a 1s electron of the cobalt atom is within the nucleus. The radius of the Co nucleus is about 4.5×10^{-15}m. The wave function for the 1s orbital is, in good approximation

$$\psi(r) = \pi^{-1/2}(Z/a_0)^{3/2}e^{-Zr/ao}$$

where $a_0 = 0.52918 \times 10^{-10}$m, and Z is the atomic number (27 for Co).

The approximate formula

$$\int_0^R r^n e^{-Cr} dr = R^{n+1}/(n+1) - CR^{n+2}/(n+2)$$

may be useful. It is valid for $R << \frac{1}{C}$.

Do not forget that the volume element for spherical integration is $dV = 4\pi r^2 dr$.

(b) Describe the principle of Mössbauer spectroscopy.

Evaluation: 80% (a)
20% (b)